

TECHNICAL PROVISIONS

SECTION 01 - TRENCH EXCAVATION & BACKFILL FOR PIPELINES AND APPURTENANT STRUCTURES

TP-101 SCOPE:

Excavation, as used in these specifications refers to all construction activities necessary to install subsurface utilities in accordance with the plans and specifications. Such activities include, but are not limited to:

All necessary clearing, grubbing and site preparation; removal of all materials that may interfere with construction activities (except existing pipe work, conduits, utility structures or other items to be left in place) to the lines and grades indicated on the plans and otherwise described herein. The Contractor shall remove all construction trash from the site and transport to a legal disposal site.

Removal and/or storage of subsurface materials from trench and construction excavation areas to allow installation of designated utilities or structures. All suitable material removed from excavations shall be used, insofar as practicable, in the formation of embankments, fills and backfilling.

Preparation of sub-grades and backfilling of trench and construction areas upon completion of utility or structure construction.

All necessary bracing, shoring and protection (but not including tight sheeting in trenches and structure excavation ordered left in place by the Owner or Owner's Representative).

Final grading, dressing and cleanup of the construction site.

TP-102 SAFETY:

All trench excavation shall be coordinated in strict accordance with current OSHA requirements found in the Occupational Safety and Health Standards - Construction Standards for Excavations (29 CFR 1926, Subpart P).

Trenches shall be properly sheeted, shored, or sloped in accordance with the current OSHA standards. Trench excavation shall not proceed more than 500 feet in advance of pipeline work without the Owner or Owner's Representative's approval. All trenches shall be completely backfilled at the end of each working day, unless otherwise approved by the Owner or Owner's Representative. No excavation shall be left open without proper barricades and warning lights. Such devices shall conform to the Manual of Uniform Traffic Control Devices (ANSI D6.1) or such permits as are appended to these specifications.

The contractor shall be responsible for safety on the job site and shall designate a trained "competent person" knowledgeable in trench safety to supervise the work.

TP-103 SHORING AND SHEETING SYSTEMS:

Protection of employees in excavations shall conform to applicable OSHA Standards. Any trench protection and modification to trenching safety plans shall be submitted to the Owner or Owner's Representative in writing to be maintained as part of the record.

The Contractor shall install all shoring and sheeting systems required to prevent cave-ins and protect his employees and adjacent property and structures in accordance with current OSHA standards. No extra payment will be made for these items, the cost thereof being merged with and considered a part of the cost for the related excavation.

Before sheeting is withdrawn, or trench boxes moved forward, they shall be raised, in place, just above the pipe crown to safely allow the Contractor to completely fill any voids left in the pipe zone.

TP-104 ROAD, RAILROAD AND SPECIAL UTILITY CROSSINGS (IF REQUIRED):

The Contractor shall be responsible for compliance with all requirements of special crossing permits applicable to this project. Copies of such permits shall be included in the Appendix of these specifications. If no special crossing permits are appended, and such crossings are indicated on the plans, crossings will comply with all applicable provisions of Section 11 of the Technical Provisions in addition to those indicated under other provisions of this Section. At least two days notice shall be given to the Owner or Owner's Representative before work is done on any crossing.

TP-105 WORK WITHIN RIGHT-OF-WAYS & TRAFFIC CONTROL

When performing any work within the right-of-way of roads or railroads, the Contractor shall comply with the right-of-way permit for the installation including all of the requirements for traffic control and compaction. All work within the right-of-way of roads shall be performed in accordance with the "Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects FP-03 U.S. Customary Units – Section 614." All work within the right-of-way of roads shall be performed in accordance with the latest edition of the Manual on Uniformed Traffic Control Devices (MUTCD)." In addition, the Contractor shall submit a traffic control plan to the project engineer for review and approval prior to any work within the right-of-way of any roads or railroads.

TP-106 DRAINAGE:

The Contractor shall control the grading in the vicinity of the excavation so that the ground surface is properly sloped to prevent water from running into the excavated areas. Water that has accumulated in the excavation from rainfall and/or surface runoff, or from any other cause which might have been prevented by proper care and foresight, shall be removed and the subgrade restored to its proper bearing capacity, all at the Contractor's expense.

TP-107 EXCAVATION:

A. General: All excavation, other than by drilling and blasting, undertaken with the excavation equipment commonly used in the industry for this type of excavated material shall be classified as common excavation.

All excavation shall be made by open cut method except as approved or specified. During excavation, materials suitable for backfill shall be neatly piled no closer than 24-inches from the edge of the excavation. All materials not required or not suitable for backfill shall be removed and wasted at locations designated by the Owner or Owner's Representative.

1. Width: The sides of all trenches for the installation of utility piping systems shall be as nearly vertical as soil conditions will permit from ground level to the pipe. Except for the trenching of 1-inch water service lines, the width of the trench shall not be less than 16-inches nor more than 24-inches wider than the outside diameter of the pipe barrel. Trench excavation shall be centered on pipe alignment such that a minimum clear space of 8-inches is provided on each side of the pipe. Trench width above the level of the top of the pipe may be as wide as necessary for shoring or sheathing and for proper installation of the work.

2. Depth: The trench shall be excavated to the depth that permits pipe to be laid at the elevations shown on the engineering drawings or with the required depth of cover specified by the Owner or Owner's Representative. Depth of cover shall be measured from the finished grade or the surface of the permanent improvement to the top of the pipe barrel.

3. Preparation: The bottom of the trenches shall be accurately shaped to line and grade and shall provide uniform bearing and support for each section of the pipe on specifically placed bedding material at every point along its entire length. Bell holes and depressions for joints shall be dug after the trench bottom has been graded and shall be only of such length, depth and width as required for properly making the particular type joint. Care shall be taken not to excavate below the depths indicated. Unauthorized over depths shall be backfilled with suitable bedding material at the Contractor's expense.

4. Previous Excavation: If the trench passes over a sewer or other previous excavation, the trench bottom shall (1) be compacted to provide support equal to that of the undisturbed native soil or (2) conform to the specific regulatory requirements that preclude damage to the existing installed facility.

5. Unstable Subgrade: Where soft, spongy or otherwise unsuitable material is encountered, which will not provide a firm foundation for pipe, the Owner will direct the extent to which removal and replacement shall be made with suitable material. Special pipe foundation material is NOT anticipated. However, if required, a price shall be negotiated between the Owner and Contractor for special pipe foundation material.

6. Underground Obstructions: The Contractor shall preserve intact any underground pipes, culverts or other utilities encountered during construction (except as hereinafter permitted) provided their location is such that they do not interfere with new pipelines or structures being installed. The Contractor shall notify all appropriate utility authorities of his construction schedule so they may be at the site to locate and protect their property. If any utilities or structures are accidentally broken or disturbed, they shall be replaced immediately to a condition at least equal to that in which they were found, all at the Contractor's expense. Couplings used to repair water service lines shall be brass compression couplings and couplings used to repair sewer service lines shall be solid sleeve couplings that provide a rigid connection between pipes. The repair work shall be done in a manner acceptable to the Owner or Owner's Representative and the utility company. Any existing water or sewer services that will intersect or interfere with the new pipelines or structures shall be rerouted by the Contractor in the manner indicated by the Owner or Owner's Representative. Existing water or sewer services from the mains to private property that interfere with trenching operations may be cut and replaced at the Contractor's option and expense, provided that users of such services are notified at least 2 hours in advance and that the use of such service shall in no case be interrupted for more than 4 hours and/or beyond the workday, unless specifically permitted in writing by the owner. Materials and construction for these items shall be as provided in other sections of these specifications. All new and existing water and sewer mains and water and sewer services shall be protected from freezing at all times during construction.

B. Rock: The inclusion of a bid item and estimated quantity for rock excavation in the bid schedule indicates that rock excavation is probable. However, the exclusion of this item from the bid schedule does not preclude the possibility that rock will be encountered; it merely indicates that it is not anticipated. If unanticipated rock is encountered, the Owner or Owner's Representative will negotiate a price for rock excavation with the contractor. The following paragraphs define solid rock and loose rock excavation.

Solid rock shall be defined as large masses of igneous, metamorphic, or sedimentary rock that, in the opinion of the Owner or Owner's Representative, cannot be

excavated without drilling, blasting, or the use of rippers or other specialized equipment. Any material excavated without the use of blasting or specialized ripping equipment shall not be considered rock.

Solid rock excavation shall be measured in cubic yards from the top of the rock to a point 4-inches below the invert of the installed pipe and an assumed 24-inches trench width, regardless of the actual trench width and depth excavated. For structures, the rock shall be profiled 12-inches outside the perimeter of the structure and computed based on a product of the profile of the rock and an assumed 24-inches outside the structure's perimeter and 6-inches below the structure's footing. The rock volume will be computed as the product of the profiled rock area, as measured by the Owner or Owner's Representative. The measurements shall be within the nearest 0.1-feet from the surface and no less than every 10-feet by one of the following methods:

1. Excavating and exposing the rock profile for measurement. This shall be the responsibility of the contractor and no additional payment shall be made for this excavation.
2. Rock profile determined by drilling without excavating and measurements taken prior to any blasting.
3. Rock profile measured after blasting and excavation. A 20% deduction shall be made in rock determination when this method is used to allow for expansion in ledge due to blasting.

Loose rock shall be defined as boulders and other detached stones each having a volume of 1 cubic yard or more. Loose rock shall be removed from the excavation in such a way that a clear distance of at least 4-inches exists between the rock and the bottom of the pipe and 6-inches exist between the rock and the bottom structure. Loose rock shall not be used for backfill. Loose rock excavation shall be measured in cubic yards, including the total volume of only those rocks or boulders that are individually over 1 cubic yard in volume.

Trench in which rock is encountered shall be excavated at least 4-inches deeper than the pipe invert and refilled to the required elevation with sand, gravel, or crushed rock passing a 3/4-inch mesh screen. Bedding material shall extend upward at least 12-inches above the pipe. Payment for this fill material shall be considered incidental to the rock excavation and no additional payment shall be made.

Should rock excavation be encountered, it shall be the responsibility of the Contractor to have an experienced powderman handle all blasting and be able to furnish proof of his/her credentials. The Contractor shall also inform all residents in the vicinity of proposed blasting activities and shall be responsible for any damage to persons or property as covered in the General Provisions.

C. Removal of Water: The contractor shall remove and dispose of all water entering the trenches and shall keep the trenches water free until the facilities are in

place and sealed against the entrance of water. Use of a trash pump for removal of nuisance water shall be at no extra cost and shall not be considered dewatering. In no case shall water, earth, or any foreign materials be allowed to enter the water main or sewer lines.

The removal of nuisance water is determined by pumping the water out of the trench with a heavy-duty 4" construction trash pump with a strainer for a minimum of 1 hour. The strainer shall be placed in a bed of pea gravel or a slotted PVC pipe in order to screen the debris.

All water removed from trenches shall be conveyed to natural drainage channels, storm sewers, or proper reservoirs as approved by Owner or Owner's Representative. Such removal of water shall be in a manner that prevents property damage, erosion, or sedimentation.

The inclusion of a bid item and estimated quantity for dewatering in the bid schedule indicates that dewatering is probable. However, the exclusion of this item from the bid schedule does not preclude the possibility that water will not be encountered, it merely indicates that it is not anticipated.

If continuous pumping with well points is required to maintain a satisfactory trench, and the contractor is so directed by the Owner or Owner's Representative, this work shall be considered as dewatering. Well points shall be set separately for each trench being dewatered. Dewatering shall be based on the actual number of lineal feet of trench dewatered and paid for at the negotiated price between the contractor and Owner or Owner's Representative.

D. Structural Excavation: Excavation for structures shall extend a sufficient distance from walls and footings to provide for forming, except where concrete for walls or footings is authorized to be deposited directly against excavated surfaces. Care shall be taken to avoid excavating below the depths indicated in the plans. Over-excavation shall be restored to proper elevation by filling with suitable granular bedding material at the Contractor's expense.

TP-108 BACKFILLING:

A. General: Trenches shall not be backfilled until the Owner or Owner's Representative has inspected and approved the pipe installation and jointing as being in compliance with the requirements of plans and specifications.

Bedding and backfill materials to a depth of 12-inches above the pipe shall be carefully deposited in layers not more than 6-inches thick, loose measurements, wetted to optimum moisture content and hand or mechanically compacted to at least 95% of the reference density for this material as described in the specification titled "Compaction Requirements, Methods and Testing. From 12 inches above the pipe to

ground surface, the excavation material shall be placed in layers not to exceed 12-inches, mounded and left in a uniform, neat condition.

Wherever trenches have not been properly filled, or if settlement occurs, they shall be reopened to the depth required for proper compaction and refilled and re-compacted as specified above and approved by the Owner or Owner's Representative.

Compaction methods and equipment may utilize hand and mechanical tampers and rollers. The equipment and procedures proposed by the Contractor shall be subject to the approval of the Owner or Owner's Representative.

B. Materials: All backfill material shall be approved in advance of installation by the Owner or Owner's Representative. Materials shall be obtained from areas approved by the Owner or Owner's Representative.

Backfill material will not be paid for separately, but shall be considered as subsidiary to and a part of the cost for the applicable contract bid item.

1. Embedment: Embedment is that material from the bottom of the trench to twelve inches above the pipe, and includes the pipe bedding material (upon which pipe is laid), haunching material (extending from pipe bottom to pipe centerline), and initial backfill material (extending from pipe centerline to 12 inches above pipe). Native soil used for embedment must be free from clods of earth or stones larger than 1 inch in any dimension, organic refuse, debris, frozen soil, and other objectionable material. If the native soil does not meet these criteria, the Contractor shall screen it (as applicable) or import special bedding material.

2. Imported Special Bedding Material: If required, special bedding material shall consist of sand, sandy gravel, or other suitable granular material having a maximum plasticity index of 6, with 100% of the bedding material smaller than 3/4-inches, and no more than 5% passing a No. 200 sieve.

3. Stabilization: Granular stabilization material shall be used to replace soft, spongy, or other unsuitable material, including rock encountered in excavation, to the depths necessary to support the pipe or structure. Stabilization materials shall be underlay bedding material (as applicable) and shall consist of suitable hard, durable granular material having a maximum size of 6-inches, graded so that a maximum of 20% passes a No. 4 sieve. Granular stabilization is not anticipated. If required, a price for granular stabilization shall be negotiated between the Contractor and the Owner.

4. Final Backfill: In general, final backfill will be that material originally excavated from the trench and will extend from 12 inches above the pipe to surface grade. Final backfill material shall be the same as that around the pipe except that the inclusion of a limited amount of stones up to 6-inches in diameter will be permitted.

C. Placement:

1. Embedment: Embedment shall be placed in 6-inch loose lifts and compacted as described herein. If over-excavation is required, bedding material is to be compacted to 95% of the maximum dry density as determined by the Standard Proctor density test (ASTM D-698). Haunching material shall be placed by hand and worked under the pipe haunch to provide adequate side support for the pipe. Haunching and initial backfill material shall be compacted to 95% of the maximum dry density as determined by the Standard Proctor density test (ASTM D-698).
2. Final Backfill: Final backfill shall not be placed until the embedment material is placed and compacted, and the Owner or Owners Representative have inspected and approved the installation. Final backfill shall be placed in lifts not to exceed 12-inches unless otherwise approved by the Owner or Owners Representative. Compaction shall be as defined in the Compaction Requirements, Methods, and Testing section.
3. Backfill for Road Subgrade: Under existing and proposed roadways, to a distance of 10-feet on either side of the road, bedding and backfill materials shall be carefully deposited in layers not more than 6-inches thick, loose measurements, wetted to optimum moisture content and mechanically compacted as described in the Compaction Requirements, Methods, and Testing section.
4. In areas where pavement is to be replaced, or in roads that are to be paved, no rocks or stones that will interfere with subgrade preparation shall be included in the backfill within 12-inches of the finished subgrade elevation. The upper 12-inch layer, forming the subgrade for pavements, shall be compacted to a density of at least 95% (ASTM D-698 - Modified Proctor Test). See Section 11 of the Technical Provisions where this is required.
5. Cement slurry can be substituted for compacted native backfill and subgrade if approved by Owner or Owner's Representative. The cement slurry shall consist of one sack of cement to one cubic yard of concrete sand and shall be placed from the concrete truck at a slump of 6 to 8 inches. Steel plates 5/8" thick are to be placed over the trench with at least 6 inches overlap on each side and edged with asphalt to prevent traffic movement. The backfill shall be allowed to set for a minimum of 12 hours before completing the asphalt patch. Slurry can typically be installed from the trench bottom to ground surface and no intermediary subgrade material is required for placement of asphalt patch.
6. Where trenches cross roads, streets, or driveways, backfilling shall be completed immediately following excavation and inspection. No trenches across roads shall remain open overnight. All crossings shall be backfilled, compacted and open to traffic at the end of each day's work. Major road crossings shall be excavated and backfilled in half widths of the traveled way so that at least one-half of the roadway is open to controlled traffic at all times during the work.

7. **Backfill Around Structures:** Backfill around structures shall conform to the same requirements as those for backfill around piping in unpaved areas, unless more stringent requirements are indicated in other sections of these specifications.

TP-109 COMPACTION REQUIREMENTS, METHODS, AND TESTING:

A. **Minimum Density:** Unless otherwise specified by applicable permits initial and final backfill and gravel resurfacing shall be compacted to the following minimum requirements. The minimum acceptable percent of compaction is the in place dry density divided by the reference density times 100. Compacted soil shall also be at plus or minus 2% of optimum moisture content.

TYPE	LOCATION	REQUIRED COMPACTION
I	Under any existing or proposed pavement, curb Gutter, sidewalk, roadway, shoulder, alley, slab, Footing, canal embankment, or when within 2 feet of above.	95%
II	Within any gas, electric, or telephone utility easement, Or within any street or road right-of-way outside the limits defined above as Type I.	90%
III	All other locations not defined above as Type I or Type II.	80% (or 100% of adjacent natural ground)

B. **Reference Densities/Baseline Testing:** The Contractor, at his expense, shall provide the reference densities for the various bedding and backfill materials used. All tests shall be performed by a certified soils testing laboratory approved by the Owner or Owner's Representative. If reference to natural ground is used, a nuclear gauge may be used to measure the density of the natural ground. The reference densities for compaction tests shall be established in accordance with ASTM D-698, Modified Proctor Test. The Contractor shall submit for approval a testing plan identifying proposed testing locations prior to the start of any excavation work.

Contractor shall provide copies of the Modified Proctor Tests with 5 point minimum moisture vs. density curves.

The contractor shall coordinate the collection of soil samples for proctor testing with the IHS construction inspector such that both parties are on-site during the collection of soil samples. This will ensure that enough samples are collected to provide for accurate density testing during construction by providing reference density for differing soil conditions within the project area.

C. Methods: Mechanical compaction is permitted. Water jetting methods are not permitted. The backfill shall be uniformly moistened to plus or minus 2% of optimum moisture content, placed in sufficiently thin layers to obtain the specified results, and compacted with hand and/or pneumatic tamp, roller, hydrohammer, or other device which will obtain the specified density without injury to the pipe or related structures.

D. Density Tests: Backfill density tests shall be performed in accordance with the latest versions of ASTM D-1556 (Sand Cone Method), ASTM D-2167 (Rubber Balloon Method), ASTM D-2216 (Moisture Content), ASTM D-2922 (Nuclear Density), and ASTM D-3017 (Nuclear Moisture Content). The Contractor will perform initial field density tests for each location listed in the next paragraph at the expense of the Contractor. Any additional tests due to failure of initial tests shall be at the expense of the Contractor.

The Contractor will perform at least one (1) compaction test every other lift at each two hundred and fifty (250) linear feet of Type I location. One compaction test will be performed every other lift on each five hundred (500) linear feet of trench at Type II or III locations. A minimum of one (1) compaction test shall be performed under each lift station base and a minimum of four (4) compaction tests shall be performed on each lift of material under proposed foundations or tanks. The exact test locations shall be specified by the Owner or Owner's Representative. The Owner may perform additional tests. If the results of any of the compaction tests indicate insufficient compaction, the area in question shall be reopened to a depth required for proper compacting, then refilled, compacted and retested, at the expense of the Contractor, until the compaction tests indicate that the necessary compaction requirements have been met. Two copies of the test results of any retesting performed by the Contractor shall be provided to the Owner, for approval, prior to any permanent surfacing. Any improperly placed backfill, or locations where settlement occurs, shall be reopened to the depth required for proper compaction, then refilled and compacted at the expense of the Contractor. The surface shall be restored and resurfaced, if necessary to the required grade.

TP-110 DISPOSAL OF EXCESS MATERIAL:

Excess material, including rock, broken concrete, bituminous materials, debris or other materials not suitable for backfill, shall be removed from the site and wasted in the disposal areas selected by the Contractor and approved by the Owner or Owner's Representative.

The disposal of such excess materials will not be paid for separately, but shall be considered as incidental to and a part of the cost for the applicable contract bid item.

TP-111 CLEANUP:

Upon completion of the work, the entire site shall be cleared of all debris, and ground surfaces shall be finished to smooth, uniform slopes and shall present neat and workmanlike appearance. All slopes shall be trimmed and dressed, and all surfaces graded such that effective drainage is assured.

Unpaved streets shall be graded smooth to the satisfaction of the Owner or Owner's Representative.

TP-112 TRENCH MAINTENANCE:

The Contractor shall, for a period of one year after completion and final acceptance of the work, maintain and repair any trench settlement that may occur and shall make suitable repairs to any pipe, pavement, or other structures that may become damaged as a result of backfill settlement.

TP-113 STORM WATER POLLUTION PREVENTION PLAN (SWPPP):

The Contractor shall prepare a Storm Water Pollution Prevention Plan (SWPPP) in accordance with the latest requirements of the Environmental Protection Agency's (EPA) National Pollutant Discharge Elimination System (NPDES) General Permit for Discharges from Large and Small Construction Activities. The SWPPP must be prepared in accordance with good engineering practices and must 1) Identify all potential sources of pollution which may reasonably be expected to affect the quality of storm water discharges from the construction site; 2) Describe practices to be used to reduce pollutants in storm water discharges from construction site; 3) Assure compliance with the terms and conditions of the NPDES General Permit.

If the Contractor is not experienced in the preparation of SWPPP, the contractor shall retain the services of an environmental firm regularly engaged in the preparation of

SWPPP to perform said service. The completed SWPPP must be approved by the Owner or Owner's Representative at least 14 calendar days before the start of construction so that a Notice of Intent can be sent to EPA.

The Contractor shall fully implement the SWPPP from the commencement of construction until final stabilization, as defined in the NPDES General Permit is achieved.

The Contractor shall maintain and update the SWPPP, as required in the NPDES General Permit, for the life of the project. Updates shall include amendments required as a result of the ineffective controls discovered through the course of inspections or investigations conducted by the Owner or Owner's Representative, site staff, or by local, state, tribal or federal officials. The Contractor shall submit a Notice of Intent to EPA to obtain permit coverage, modify the coverage as necessary, and terminate permit coverage once final stabilization is achieved.

TP-114 LINES AND GRADES:

The Engineer will give all lines, grades and building locations on the plans and will supply the contractor with the AutoCAD drawing to stake out the facilities to be installed. The CONTRACTOR shall be responsible for staking out pipeline centerlines with a lath every 200 feet or line-of-sight whichever is less. Bends, intersections, manholes, lift station centers and fence corners shall be staked by the CONTRACTOR and provided with two offsets for alignment. Elevation references will be provided as shown on the plans, at lift station and for sewer manholes. The Contractor shall be responsible for the preservation of the location and line and grade stakes when set, and if disturbed, shall have such stakes replaced.

TP-115 CLEARING & GRUBBING:

It is the contractor's responsibility to clear and grub the site prior to or during construction. The contractor shall remove all trees along the water and sewer main alignments. Trees may either be chipped with a woodchipper and placed over the trench for erosion control or disposed of at the Contractor's expense. Clearing and grubbing shall be done at the contractor's expense.

TP-116 SEEDING:

All disturbed areas shall be returned to their pre-construction vegetative state. The contractor shall submit a seed mix that is equivalent to the native vegetation in the

area of construction. The contractor shall protect the seed after it is placed with hay mulch, straw mulch, wood cellulose mulch, or as approved by the project engineer. A minimum of 20 pounds of seed per acre shall be placed. Seed shall be placed by either drill seeding at a depth of approximately 1 inch or broadcast seeding. If broadcast seeding is utilized, the contractor shall apply twice the minimum seeding rate (40 pounds of seed per acre). The contractor shall perform maintenance as needed to ensure that adequate vegetative growth and stabilization has taken place to minimize erosion after construction is completed.

TP-117 FROST PROTECTION:

- A. Materials: The insulation shall be rigid extruded polystyrene insulation board, having a minimum compressive strength of 25 psi. The width shall be 2 feet for mains, service lines and effluent discharge lines less than 6-inches (nominal diameter). The thickness shall be 2-inches.
- B. Placement: When indicated on the drawings and in the bid schedule, insulation shall be placed in areas where water lines, sewer lines, or effluent discharge lines are susceptible to freezing. The insulation shall be centered over the main with no more than 6 inches of compacted fill between the pipe and insulation. The Contractor shall grade fill so insulation lays flat and maintain a straight alignment of insulation. The Contractor shall lap insulation by 6 inches or stagger by 6 inches if composed of two layers. The thickness for the first lift of backfill over the insulation shall be a minimum of 8 inches. The Contractor shall not operate construction equipment directly on insulation and not compact the first lift with the backhoe-mounted compactor, or any other large compaction equipment. The remaining backfill shall be compacted using normal construction practices.

TP-118 REPAIRS TO DAMAGED UTILITIES

The contractor is responsible for repairing any utilities that they damage during construction at no cost to the Owner. Repairs shall be made in accordance with the requirements of each utility. Below are the requirements for making repairs to damaged water and sewer utilities.

Water Mains: If the damage is small a small crack or hole in the water main, the contractor shall install a stainless steel repair coupling equal to a Romac SS1, SS2, SS3 or Ford Style F1. If the damage is too large to repair with a repair coupling, the Contractor shall install new water main to replace the damaged water main. The repair must result in a watertight water main that does not leak. The new water main shall be DR-18, Class 235, C900 PVC water main per TP-4. The new water main shall be connected to the existing water main using restrained joint solid sleeve

couplings equal to Romac Alpha series restrained couplings.

Water Service Lines: If the damage is small a small crack or hole in the water service line, the contractor shall install a stainless steel repair coupling equal to a Ford Small Repair Clamp with Full Wrap Gasket FSC. If the damage is too large to repair with a repair coupling, the Contractor shall install new water service line to replace the damaged water service line. The repair must result in a watertight water service line that does not leak. The new water service line shall be 200 psi rated polyethylene water service line meeting TP-5 requirements. The new water service line shall be connected to the existing water service line using stab joint, compression joint, or pack joint brass fittings as specified in TP-5.

Sewer Mains & Service Lines: If the damage is small a small crack or hole in the sewer main or service line, the contractor shall install a stainless steel repair coupling equal to a Romac LSS1, LSS2, or LSS3. If the damage is too large to repair with a repair coupling, the Contractor shall install new sewer main or service line to replace the damaged sewer main. The repair must result in a watertight sewer main or service line that does not leak. The new sewer main shall be SDR-35 PVC sewer main meeting TP-6 requirements and the new sewer service line shall be SDR-35 PVC sewer service line meet TP-7 requirements. The new sewer main or service line shall be connected to the existing sewer main using solid sleeve couplings equal to Romac 501 sewer couplings.

TP-119 AS-BUILT DRAWINGS:

- A. General: The as-constructed drawings shall be a record of the construction as installed and completed by the Contractor. They shall include all the information shown on the Contractor's set of drawings and a record of all deviations, modifications or changes from those drawings, however minor, which were incorporated in the work, all additional work not appearing on the contract drawings and all changes which are made after final inspection of the contract work.
- B. As-Built Drawings: The Contractor shall mark up one set of paper prints to show the As-Built Drawing information. These Asbuilt Drawing prints shall be kept current and available on the job site at all times. All changes from the contract plans which are made in the work or additional information which might be uncovered in the course of construction shall be accurately and neatly recorded as they occur by means of details and notes. No construction work shall be concealed until the necessary record data has been recorded. The Asbuilt Drawing marked prints will be jointly inspected for accuracy and completeness by the Owner and a responsible representative of the construction Contractor prior to submission of each partial payment, as evidenced by the issuance of a receipt by the Owner indicating the adequacy of the information. Failure to keep the as-constructed marked prints on a

current basis shall be sufficient justification to withhold approval of request for payment or suspend pay estimates. The drawings shall show the following information, but not limited thereto.

1. The location and description of any utility lines or other installations of any kind or description known to exist within the construction area. The location includes dimensions to permanent features.
2. The location and dimensions of any changes from the contract drawings.
3. Changes in details of design or additional information obtained from working drawings specified to be prepared and/or furnished by the Contractor including but not limited to fabrication, erection, installation plans, and placing details, pipe sized, insulation materials, dimensions of equipment foundations, etc.
4. All changes or modifications which result from the final inspection.
5. All information as required in the technical provisions.

- C. Electronic/Surveyed As-built Drawings: The contractor shall obtain the services of a surveyor licensed in the State of New Mexico to survey in the locations of all sanitation facilities installed by the project being constructed. The number and location of surveyed points must be sufficient to provide enough detail to accurately identify the location of the installed sanitation facilities. Points to be surveyed shall include, but not be limited to:

Water Mains & Services: gate valves, fire hydrants, flush hydrants, water meters, air/vacuum release valves, water/sewer crossings, water main bends, water service line connection, water main intersections, casing pipe, water main reducers, and other water main components installed under this project.

Other Water Facilities: wells, booster pumps, valve vaults, building corners, fence corners, water tanks (including base and overflow elevations), and other water main components installed under this project.

Sewer Mains & Services: manholes (including rim elevation, invert in elevation, and invert out elevation), terminal sewer main cleanout rim and invert, sewer service cleanout, new sewer service line connection, and other sewer components installed under this project.

Other Sewer Facilities: lift stations (including base elevation, rim elevation, invert in elevation, lead/lag pump on elevation, all pump off elevation, and high/low alarm elevation), valve vaults, gate or plug valves, air/vacuum valves, building corners, fence corners, treatment tanks (including rim elevations, invert in elevations, and

invert out elevations), treatment units, lagoons (including floor elevation, top of berm elevation, and invert elevations of all piping in transfer structures), and other sewer components installed under this project.

The contractor shall furnish the Owner with electronic asbuilts of the facilities installed on this project in ACAD format (dwg file). The project engineer will furnish the contractor with an electronic copy of the contract drawings in ACAD format and will provide control points for use by the contractor to draft the electronic asbuilt drawings. The labeling, linework, and format of the electronic asbuilts shall be similar to that of the contract drawings.

- D. Review and Approval: One set of the preliminary As-built Drawings marked prints shall be delivered to the Owner before final inspection for his review and approval. The review by the Owner will be expedited; however, the Owner cannot guarantee to review more than one complex mechanical or electrical Record drawing sheet per working day. Upon disapproval of the As-built Drawings one set of marked prints will be returned to the Contractor for further work and resubmitted to the Owner.
- E. Other: All costs incurred by the Contractor in the preparation and furnishing As-built Drawings shall be included in the contract price and no separate payment will be made for this work.

TP-119 MEASUREMENT AND PAYMENT:

- A. General: Except for the following items, the cost of all work done by the Contractor as required under Section 01 of the Technical Provisions shall be merged with the pay items defined within the Measurement and Payment portions of other Sections of this contract.
- B. Rock Excavation: Payment for rock excavation shall be at the unit price listed in the Bid Schedule based on the computed number of cubic yards removed. No differentiation payment will be made between solid or loose rock excavations.
- C. Dewatering: Dewatering shall be based on the actual number of lineal feet completed. Payment for dewatering shall be at the contract unit price shown in the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for a complete dewatering installation.
- D. Mobilization/Demobilization: Payment for mobilization/demobilization shall be at the unit price listed in the bid schedule. 60% of this line item may be requested upon complete mobilization to the job site and the remainder may be requested upon demobilization from the job site.

E. Storm Water Pollution Prevention Plan: Payment for the preparation and implementation of the SWPPP shall be paid on a lump sum basis as shown on the Bid Schedule. Payment shall be full compensation for plan preparation including required revisions for Owner's acceptance, updates to the SWPPP for the life of the project, permit application, inspections, installation and maintenance of controls, modification of controls as determined by inspections, removal of pollutants due to failed controls, and permit termination.

F. Seeding: Seeding shall be paid for on a lump sum basis to seed the site in accordance with these specifications. Payment for seeding shall be at the contract unit price shown in the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for complete installation.

G. Exploratory Time: Exploratory time shall be measured on an hourly basis for an actual period spent on locating the existing utility line exceeding two hours. Contractor shall follow these steps:

1. Call the representative from the operating Utility and make every effort to locate the existing utility line prior to excavation.
2. Locate the existing utility line for two hours at the Contractor's expense.
3. If the Contractor is unable to locate the existing utility line within two hours, the Contractor shall notify the Owner or Owner's Representative and both agree upon a start time. The start time shall be recorded. When the Contractor locates the existing utility line, the end time shall be recorded.

If the Contractor fails to notify the Owner or Owner's Representative when the Contractor will start locating the existing utility line, the Contractor will not be compensated. Payment for exploratory time shall be at the contract unit price shown in the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for locating the existing utility line.

H. Frost Protection: Payment for frost protection shall be based on the actual number of lineal feet completed. Payment for frost protection shall be at the contract unit price shown in the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for a complete installation.

TP-120 APPROVALS:

- A. Stabilization material, if required
- B. Bedding material, if required

- C. Rock excavation method, if required
- D. Dewatering procedures, if required
- E. Baseline Proctor density test results (5 point moisture density curves)
- F. Baseline testing location plan
- G. Soil Testing Lab Certification
- H. Stormwater Pollution Prevention Plan, if required
- I. Blasting contractor and credentials, if required
- J. Seed Mix
- K. Insulation, if required
- L. Traffic control plan, if required

TECHNICAL PROVISIONS

SECTION 06 – GRAVITY SANITARY SEWERS

TP-601 SCOPE:

The work covered by this section consists of furnishing all plant, labor, equipment, materials and incidentals, in connection with the construction of gravity sewer mains and appurtenances, in accordance with these drawings and specifications.

TP-602 GENERAL:

All facilities shall be constructed in the locations to the grades and of the sizes shown on the drawings. Excavation, trenching and backfilling shall be in accordance with Section 01 of these Technical Provisions. Any section of sewer that is found defective in material, alignment, grade or joint shall be corrected so as to meet these specifications and drawings.

TP-603 MATERIALS:

A. Pipe, Joints and Fittings

1. PVC Gravity Sewer Pipe: Sewer pipe shall be PVC and shall conform to all requirements of product standard ASTM D3034 for pipe diameters up to 15 inches and ASTM F679 for 18-inch to 48-inch diameter sewer mains. Sewer pipe shall meet the pipe compound requirements of ASTM D1784 and shall be made with PVC 1120 resin, Type I, Grade I. Pipe stiffness shall meet ASTM D2412. Pipe shall be nominal size, SDR-35. Joints shall be furnished with one end belled. The joint shall be integral bell and spigot with a Rieber rubber gasket. The integral bell shall meet ASTM D 3212. The gaskets shall be as recommended by the pipe manufacturer and shall meet ASTM F477.

Plastic pipe with scratches, gouges, or grooves deeper than 10 percent of the wall thickness shall be rejected.

2. Services Wyes and Risers: PVC fittings utilized shall be single gasketed bell and spigot push-on type, meeting ASTM 3034.
3. Ductile Iron Pipe: All ductile iron pipe shall be in accordance with AWWA C151 and shall be in 18 to 20 foot lengths with single rubber gasket (push-on) joints in accordance with AWWA C111. Pipe shall be 350 psi rated. All ductile iron pipe used for sewer shall be cement-mortar lined in accordance with ASTM A746 and AWWA C104.

B. Manholes

1. General: All concrete used in constructing the manhole shall conform to Section 02 of the specifications. All concrete used for manholes shall be batched with a chemical resistant admixture for protection against hydrogen sulfide gas. The admixture shall be equal to Moxie 1800 Super-Admix as manufactured by Moxie International. Contractor shall provide most recent concrete cylinder break result which shall be no more than 30 days old. . Manholes shall meet the minimum standards of ASTM C478-18 – Standard Specification of Circular Precast Reinforced Concrete Manhole Sections.
2. Lines and Grades: All manhole locations shall be as shown on the drawings unless field changes are necessary. Sewer line cut stakes shall be provided by the Contractor at each manhole and at the midpoint between each manhole at a minimum. A minimum of 2 offset stakes shall be provided for each cut stake. Further information on staking is found in TP-1.
3. Manhole Frames and Covers: Frames and covers shall be made of cast iron with a combined weight (frames and cover) of not less than 266 pounds. Cast iron frames and covers shall conform to drawings in all essentials of design. The cover shall have a lifting pocket.. Manhole frames and covers shall meet ASTM A48. Before leaving the foundry, all castings shall be thoroughly cleaned after which they shall be dipped twice in a preparation of hot asphalt or coal tar and oil, in such a manner as to form a firm coating. The manhole and covers shall be set such that the top of the cover shall be 3 inches above unpaved roads where paving is expected later or ground level as shown on the drawings or as directed by the Owner or Owner's. Frames and covers shall be East Jordan Iron Works, Inc. Product No. 00120211, 1202Z Frame with vent hole (1202 Assembly, Product No. 00120202) or without vent hole (1202A Cover, Product No. 00120263) or approved equal.
4. Manhole Bases: Bottom manhole sections shall have integral precast base or floor slabs. Manhole bases shall be a minimum of 6 inches thick for 48 inch diameter manholes and a minimum of 8 inches thick for manholes larger than 48 inches. If precast concrete manhole bases are used, they shall be bedded on 8 inches of crushed rock or gravel meeting the following requirements: 100% passing a 40 mm screen and not more than 10% passing a No.40 sieve, with uniform grading. Precast bases with precast channels shall be pre-approved on a case-by-case basis by the Owner or Owner's representative. Cast in place base slabs shall be constructed with reinforced concrete as shown in the detail plans. The contractor shall submit detail drawings showing the size, placement, and spacing of reinforcing bars for manhole bases. This requirement applies for both pre-cast and cast-in-place manhole bases. Cast in place base slabs shall be constructed with a keyed joint for the

manhole barrels to be installed into as shown on the detail drawings.

5. Standard Manholes: Standard manholes are those which have a depth, measured from the invert of the outlet pipe to the top of the cover, greater than 6 feet. Standard manholes shall be constructed of 4 feet inside diameter, 5 inches thick precast concrete manhole sections. Cone sections shall be eccentric, 5 inches thick precast concrete. Standard manholes shall have a 6 inch thick concrete base slab, and shall be constructed as shown on the detail drawings.
6. Shallow Manholes: Shallow manholes shall have a depth of 6 feet or less, and shall be constructed of 4 feet inside diameter, 5 inches thick precast concrete sections with 8 inch thick reinforced concrete flat top cover and 6 inch thick base slabs. Shallow manholes shall be constructed as shown on the detail drawings.
7. Inlet Manholes: Inlet manholes shall have a depth of 6 feet or less, and shall be constructed of 4 feet inside diameter, 5 inches thick precast concrete sections with reinforced concrete cover 8 inches thick and base slabs 6 inches thick, respectively in accordance with ASTM C478-18. The inlet manhole shall be constructed as shown on the detail drawings.
8. Diversion Manholes: Diversion manholes shall have a depth of 6 feet or less, and shall be constructed of 4 feet inside diameter, 5 inches thick precast concrete sections with reinforced concrete cover 8 inches thick and base slabs 6 inches thick, respectively in accordance with ASTM C478-18.. Diversion manholes shall be constructed as shown on the detail drawings. Each diversion manhole shall be provided with a fiberglass gate frame and gate. The fiberglass gate frame shall have a thickness of 3/16-inch. The gate frame shall have a channel that is 1/4-inch. The fiberglass gate shall be 1/4-inch thick. The gate frame and gate shall be constructed as shown on the detail drawings.
9. Drop Manholes: An interior drop bowl, drop pipe, and fasteners shall be installed for the inlet pipe in the existing drop manhole. The interior drop bowl shall be B-8 Drop Bowl, part number B8DB, as manufactured by Reliner/Duran, Inc. of Lyme, Connecticut or equal. The interior drop bowl shall be installed as per manufacturer's recommendation using stainless steel fasteners. The drop pipe shall be securely attached to the manhole wall using stainless steel Reliner Adjustable Clamping Brackets or equal and stainless steel fasteners. The drop pipe shall be 8-inch SDR35. The adjustable clamping brackets shall be made of 1-1/2" width, 11 gauge, type 304 stainless steel. The pinch bolts and nuts shall be 3/8" diameter 18-8 stainless. Bracket interval shall be 4 feet maximum and a minimum of two brackets. The connection of the drop bowl to the drop pipe shall be by flexible external pipe

coupler. The turn-out at the base and end of the drop pipe shall be installed using a 45 degree PVC pipe elbow.

10. Waterstops: All pipe penetrations through the manhole shall be equipped with waterstop devices. Waterstop devices shall be equal to concrete manhole adapter as manufactured by Fernco. Grouting shall be only applied between the penetrating PVC and receiving concrete invert to create a smooth, watertight transition.. Grouting shall not be applied to or around the boot connectors.

- C. Terminal Sewer Main Cleanouts: Such cleanouts shall be of the same material as the gravity sewer main and sized as called for on the bid schedule. Cleanouts are to be constructed as shown on the detail drawings. Frame and lid shall be equal to East Jordan Iron Works, Inc. Product No. 00157804, Catalog No. 1578ZPT and 1578APT with a locking cover. Covers shall be marked "SEWER. The open pipe inside the locked cover shall be plugged with a Cherne Original Gripper, or equal.
- D. Warning Tape: Warning tape shall be green in color with "Caution - Buried Sewer Line Below" continuously printed on it. Tape shall be a minimum of 3-inches wide, 5 mils total thickness, composed of plastic with a metal foil core, and equal to Traceline Underground Utility Marking Tape. The warning tape shall be installed 12 inches above the pipe. This tape shall also be installed on sewer service lines.
- E. Marker Post: 2-1/2 inch diameter aluminum or steel utility markers with stampable brass or aluminum caps shall be installed to mark the location of all manholes and terminal sewer main cleanouts. Marker posts shall be painted with a minimum of 2 coats of yellow paint designed for outdoor commercial use. The marker posts shall be properly cleaned and the surface prepared in accordance with the paint manufacturer's recommendations prior to painting. Markers shall be 60-inch long, 2.5 inch diameter, Bernsten Model A1NBF60-OS aluminum posts, or approved equal. Marker posts shall be installed, to leave 36 inches exposed above ground. Where possible, the markers shall be located immediately above the indicated item. The Owner will specify the location for markers requiring offset installation. The Contractor shall be responsible for accurately stamping the location, orientation and size of the sewer main, manholes, terminal cleanouts, and appurtenances on the cap. Installation and stamping of marker posts shall be considered incidental to installation of sewer main, manholes, terminal cleanouts, and appurtenances.

Under no circumstances shall marker posts be installed within the right-of-way of any roadways, unless specifically indicated on a right-of-way permit. Instead, the contractor shall install carsonite marker posts to mark the locations of valves, vaults, bends, tees, crosses, interconnections, etc. within the right-of-way of roadways. Carsonite markers shall be 72 inches in length, 3.75 inches in width, green in color, model CRM306607 with anchor barb and "Caution Sewer Pipeline" text on the

marker as manufactured by Carsonite Composites, or equal. The carsonite markers shall be installed directly over the item that it is marking with a 24 inch bury depth.

TP-604 TRENCH EXCAVATION AND BACKFILL:

Trenching and backfilling operations shall be performed as specified in Section 01 of the Technical Provisions.

TP-605 INSTALLATION OF GRAVITY SEWER LINES:

The bottom of the trench shall be shaped to give uniform support to the pipe. Pipe laying shall proceed upgrade, with the spigot end pointing in the direction of the flow. Each pipe shall be laid true to line and grade as shown on the drawings, and in such a manner as to form a close concentric joint with the adjoining pipe. As the work progresses, the interior of the sewer shall be cleared of all dirt and extraneous materials of every description. If the maximum width of the trench at the top of the pipe specified in Section 01 of these Technical Provisions is exceeded for any reason other than at the direction of the Owner or Owner's representative, the Contractor shall install such concrete cradling, encasement, gravel base or other bedding as may be required to satisfactorily support the added load of the backfill. Trenches shall be kept free from water and the pipe shall not be laid when conditions of the trench or weather are unsuitable for such work. At all times when work is not in progress, all open ends of pipe and fittings shall be securely closed so that no trench water, earth or other substances will enter the pipe.

TP-606 WATER AND SEWER LINE SEPARATION REQUIREMENTS:

- A. General: Water lines located near sewers present conditions for serious potential cross contaminations. Protection from cross contamination can be provided by separation of the facilities and use of special piping materials. For measuring separation between pipes, all measurements shall be the clearance between pipes (pipe O.D. to pipe O.D.).

For the purposes of this section, the term "lines" shall include mains, laterals, and service lines for both water and sewer.

- B. Horizontal Separation of Water and Sewer Lines: When water and sewer lines are laid parallel to each other, the horizontal distance between the water and sewer lines shall be at least 10 feet. Each line shall be laid in a separate trench.

When physical conditions, such as an existing obstruction, will not allow the required 10-foot horizontal separation, the water and sewer lines may be laid closer than 10 feet if the bottom of the water line is at least 18 inches above the top of the sewer line.

- C. Vertical Separation of Water and Sewer Lines: When water lines cross sewer lines, the water line should be above the sewer line with no less than 18 inches vertical clearance (Measuring pipe O.D. to pipe O.D.).

Whenever possible the water line shall be above the sewer line. If the minimum 18-inch clearance cannot be obtained with the water line above the sewer, the water line must be installed under the sewer line.

Where a water line must cross under a sewer line, a vertical clearance of at least 18 inches between the bottom of the sewer line and the top of the water line shall be maintained. The water main shall be the normal water distribution pipe as specified on the construction drawings, with a 20-foot pipe section centered on the sewer crossing. No joints in sewer lines shall be permitted within 9 feet of a water line.

Where a water line must cross sewer service lines or mains, and the water line is not a minimum of 18 inches above the sewer service line or main, special protection is required. In these instances, the sewer service line or main shall be reconstructed of ductile iron pipe of the same size as the original sewer service line or main for a distance of 10 feet on either side of the water/sewer crossing point. As an alternative method of providing special protection, the contractor may place concrete a minimum of 8" greater than the outside diameter of the pipe (i.e. 4" of concrete all the way around the pipe) for a length of 20 feet (10 feet on either side of the water/sewer crossing point), if approved by the project engineer. Concrete shall meet TP-2 requirements. All existing sewer grades shall be maintained.

- D. Water Main Separation from Sewer Manholes: No water pipe shall pass through, under, or come into contact with any part of a sewer manhole.
- E. Separation Between Water Lines and Components of the Sewage Disposal System: Water lines shall not be installed within 10 feet of a septic tank, within 25 feet of a drainfield, or 50 feet from an outhouse. Also, waterlines shall not be installed within 100 feet of the perimeter fence of an individual lagoon, or within 500 feet of the perimeter fence of a community lagoon.

TP-607 CONCRETE ENCASEMENT OF SEWER MAIN:

Sewer line sections indicated on the plans as requiring concrete encasement shall be constructed in accordance with the detail drawings. Transition from PVC to DI sewer pipe shall be accomplished with approved a transition boot. DI piping shall extend a minimum of 6-1/2 feet beyond the limits of indicated concrete encasement unless otherwise directed in the plans and detail drawings. Concrete requirements shall be as indicated in Section 02 of these Technical Provisions.

TP-608 INSTALLATION OF MANHOLES:

Manhole invert channels shall be smooth and semi-circular in shape, conforming to the inside of the

adjacent sewer pipe section. The invert shall be finished smoothly with a semi-circular cross section. Flat-bottomed inverts shall not be acceptable. Inverts with humps or low spots or roughness of finish which will catch solid materials will not be acceptable. Inverts shall not be brush finished. A minimum invert elevation drop of one-tenth of a foot from the entrance to the outlet shall be provided in all manholes where there is a change in direction, or change in grade. Changes in direction of flow shall be made with a smooth curve of as large a radius as the size of the manhole will permit. S-curves will not be acceptable. Changes in sizes and grade of the channels shall be made gradually and evenly. The manhole shall be positioned such that the pipes intersect in the center of the manhole circle. The invert channels will be formed directly in the concrete. For those manholes where the sewer pipe does not change grade or slope, the invert may be constructed by laying a full section of sewer pipe through the manhole and cutting out the top half after the surrounding concrete has hardened.

The floor of the manhole outside the channels shall be smooth and shall slope toward the channel at a 15 percent grade. , installed as shown on the drawings. Any deviation of alignment of more than 3 inches for any step shall be cause for it to be reinstalled at the Contractor's expense.

The tops of all manholes shall have a concrete collar of the dimensions shown on the detail. Manholes shall be installed at the locations and elevations shown on the site plans or as directed by the Owner or Owner's representative in the field. Joints between precast manhole sections shall be sealed with "Ram Nek" bituminous rope type sealer, Reduced Friction Seal pre-lubricated gasket or equal. The sections shall then be grouted to a smooth finish on the interior and exterior of the manhole. Grout for jointing shall be as specified in Section 02 of the Technical Provisions. Manhole sections, and adjustment rings if required, shall be grouted in place when the manhole is constructed. The grout shall be spread evenly over the entire mating surface. The maximum number of adjustment rings shall be as indicated on the plans. The jointing and sealing materials shall be approved by the Owner or Owner's representative prior to installation.

Any drop manholes shall be constructed at the locations shown on the plans in accordance with the details shown on the construction drawings. Bases, walls and cones shall conform to the drawings. The manhole covers shall be set to the elevations shown on the drawings.

All connections between sewer pipe and manhole walls shall be sealed with non-shrinking grout in such a manner to make the manholes water tight.

Manholes shall not be acceptable if any evidence of infiltration into them is found. The Contractor shall take whatever actions are necessary, at his expense, to ensure that the manholes are completely watertight.

Precast manholes shall not be altered or modified in anyway. Precast manhole bases, barrel sections, etc. shall be installed in their precast condition. Chipping or altering the concrete shall not be allowed.

TP-609 CONNECTION TO EXISTING MANHOLES:

Connection of newly constructed sewer mains to existing manholes shall be of either the drop or at-grade type as indicated in the drawings and listed on the bid schedule. Reconstruction of manhole bases and inverts, all necessary piping, and associated work required to complete the connection shall comply with the provisions of Section TP-608.

All pipe entrances into the existing manhole shall be core bored, and a waterstop and watertight sealer shall be installed on each pipe penetrating into the manhole. All voids shall be repaired with non-shrink grout. Grouted areas shall be painted with bituminous coating on exterior. The invert shall be constructed as specified in TP-608 Installation of Manholes. Chipping and altering existing concrete inverts to connect the new sewer main pipe into the manhole is allowed.

TP-610 TESTING:

- A. Sewer Main Alignment: After the gravity sewer lines have been completely backfilled and the contractor has verified the line as ready for testing, the Owner or his/her representative will check the alignment by shining a light between manholes. Any deviation from true line or grade may be cause for rejection of the line. Any deviation from true line or grade which prevents water from draining by gravity from the sewer system, including manholes, shall be corrected such that the facilities meet these specifications and drawings. For horizontal alignment only, a deviation allowance of one-quarter of the inside diameter of the pipe may be excepted by the Owner or Owner's representative.
- B. Sewer Main Water Tightness:

Testing for water tightness in gravity sewer mains may be accomplished by either of the following two methods:

- 1. Exfiltration/Infiltration: Tests for water tightness in the gravity sewer mains shall be made by the Contractor in a manner approved by and in the presence of the Owner or Owner's representative. The sewer and manhole connections shall not leak under either external or internal water pressure in excess of 10 gallons per day per inch diameter per 1000 feet of sewer. The test shall be run for a minimum of a four hour period.

Infiltration testing shall be completed under existing ground water conditions.

Exfiltration testing shall be conducted under an internal pipeline test pressure generated by a manhole water level at least 1 foot above the highest elevation of the sewer main to be tested. Leakage by either infiltration or exfiltration greater than specified above shall be corrected by the Contractor at his expense. All equipment and water for these tests shall be furnished by the Contractor. This test will be made after the line has been completely backfilled.

2. Low Pressure Air Test:

- Test Requirements & Setup:
 - a. Test each newly installed section of gravity sewer line between manholes.
 - b. Slowly introduce air pressure to approximately 5.0 psig.
 - c. Allow pressure to stabilize for at least five (5) minutes. Adjust pressure to 5.0 psig or the increased test pressure as determined below if groundwater is present. Start the test.
- Test Procedure:
 - a. Determine the test duration for a sewer section with a single pipe size from the table below:

Low Pressure Air Test – Test Times	
Sewer Main Diameter (inches)	Test Time (Minutes/100 Feet of Sewer Main)
8	1.2
10	1.5
12	1.8
24	3.6

- b. Record the drop in pressure during the test period. If the air pressure has dropped more than 1.0 psig during the test period, the line is presumed to have failed. If the 1.0 psig air pressure drop has not occurred during the test period, the test shall be discontinued and the line will be accepted.
 - c. If the line fails, determine the source of the air leakage, make corrections and retest. The Contractor has the option to test the section in incremental stages until the leaks are isolated. After the leaks are repaired, retest the entire section between manholes.
- C. Flushing: The Contractor shall flush all sewer lines before pre-final inspection to remove sand, silt and other foreign material which might have entered the pipe during construction. Water used for flushing shall be domestic quality or as approved by the Owner or his/her representative. All equipment and water for the flushing shall be furnished by the Contractor. This test will be made after the line has been completely backfilled. The Contractor shall dispose of all water and foreign matter after flushing in an approved manner.
- D. Deflection Test: The maximum allowable deflection (reduction in vertical inside diameter) for PVC pipe shall be five percent. However, up to seven and one half

percent may be allowed 30 days after final backfilling. Deflection testing is required in all cases between all manholes. All locations with excessive deflection shall be excavated and repaired by re-bedding or replacement of the pipe. Acceptable methods of deflection testing include use of properly sized go-no-go mandrels or deflectometer. Deflection testing is not required for ductile iron sewer mains.

D. Manhole Testing: All manholes shall be tested for watertightness. Each manhole shall be tested immediately after assembly and prior to backfilling. All lift holes shall be plugged with an approved non-shrink grout. All pipes entering the manhole shall be plugged, taking care to securely brace the plug. If the manhole fails the initial test, necessary repairs shall be made and the manhole shall be retested. One of the following methods shall be used.

1. Vacuum Testing: Vacuum testing should be conducted in accordance with ASTM C-1244 (Vacuum Test for Concrete Manholes), except as modified below. The vacuum test head shall be placed inside the top section and the seal inflated in accordance with the manufacturers' recommendations. A vacuum of 10 inches of mercury shall be drawn and the vacuum pump shut off. With the valves closed, the time shall be measured for the vacuum to drop to 9 inches. The manhole shall pass if the time is greater than 60 seconds for 48-inch diameter, 75 seconds for 60-inch, and 90 seconds for 72-inch diameter manholes.
2. Hydrostatic Testing: Hydrostatic testing shall be conducted in accordance with ASTM C-969 except as modified below. The mains into and out of the manhole shall be stoppered with a suitable device such as a tethered pneumatic plug. The manhole shall be filled with water to the ring. After a period of at least one hour to allow for concrete absorption and to allow the water level to stabilize, the manhole shall be refilled and the water level shall be checked. The hydrostatic test shall then begin and shall be administered for a period of 4 hours. If the water level is found to drop more than $\frac{1}{4}$ inch per foot of depth of the manhole over this 4 hour duration, then the leakage shall be considered excessive and the Contractor shall be required to make all necessary repairs and retest the manhole. The exterior of the manhole shall be inspected during this period for visible evidence of leakage. Visible moisture, sweating, or beads of water on the exterior of the manhole shall not be considered leakage, but any water running across the concrete surface will be considered leakage and shall be repaired to the satisfaction of the Engineer regardless of the volume of water lost during the test.

TP-611 CLEANUP:

Upon completion of the work, the entire site shall be cleared of all debris, and the ground surface shall be finished to smooth and uniform slopes. All fences, clotheslines, gravel driveways or other

obstructions removed during construction shall be left in a condition at least equal to their condition prior to construction. Cleanup shall be considered an incidental item and no additional payment shall be made for it, but rather its costs shall be merged with the applicable pay item regardless of whether cleanup is specifically included in the measurement and payment section.

TP-612 MEASUREMENT AND PAYMENT:

- A. Gravity Sewer Main: Gravity sewer main shall be measured in linear linear feet along the centerline of the pipe, center to center of the manhole without deduction for fittings or diameter of manholes, for each of the various sizes and types of pipe installed. Payment for sewer pipe shall be at the contract unit price for the various sizes and types of sewer main installed as shown on the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, exfiltration/infiltration testing, and incidentals required for a complete installation, including but not limited to excavation, bedding, stabilization material, laying and jointing pipe, exfiltration/infiltration testing or air testing, supplying water, plugging, measuring, flushing, backfilling, as-builts, and final cleanup.
- B. Shallow Manholes: Shallow manholes shall have a depth of 6 feet or less when measured from the invert of the outlet pipe to the top of the cover, and shall be measured each. Payment shall be at the contract unit price shown on the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for a complete installation, including but not limited to excavation, concrete, frame and cover, adjustment of height, invert forming, connection to sewer lines, backfilling, as-builts, and final cleanup.
- C. Standard Manholes: Standard manholes shall have a depth greater than 6 feet when measured from the invert of the outlet pipe to the top of the cover, and shall be measured each. Payment shall be at the contract unit price for the various depths of manholes installed as shown on the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete installation, including but not limited to excavation, concrete, frame and cover, adjustment of height, invert forming, connection to sewer lines, backfilling, as-builts, and final cleanup.
- D. Drop Manholes: Drop manholes shall be measured each. Payment shall be at the contract unit price shown on the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete installation, including but not limited to excavation, concrete, frame and cover, adjustment of height, invert forming, connection to sewer line, backfilling, as-builts, and final cleanup.
- E. Inlet Manholes: Inlet manholes shall be measured each. Payment shall be at the contract unit price shown on the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete

installation, including but not limited to excavation, concrete, frame and cover, gate frame and gates, adjustment of height, invert forming, piping, connection to sewer line, backfilling, as-builts, and final cleanup.

- F. Diversion Manholes: Inlet manholes shall be measured each. Payment shall be at the contract unit price shown on the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete installation, including but not limited to excavation, concrete, frame and cover, gate frame and gates, adjustment of height, invert forming, connection to sewer line, backfilling, as-builts, and final cleanup.
- G. Terminal Sewer Main Cleanouts: Sewer main cleanouts installed at the terminal end of a sewer main shall be measured each. Payment shall be at the contract unit price on the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete installation, including but not limited to excavation, concrete, frame and cover, fittings, backfilling, as-builts, and final cleanup.
- H. Connection to Existing Manhole: Connections to existing manholes shall be measured each. Payment shall be at the contract unit price shown in the Bid Schedule. This price shall be full compensation for furnishing all labor, equipment, material, and incidentals required for a complete installation, including but not limited to excavation, cutting into the existing manhole, grouting, fittings, removing the existing invert, pouring and forming a new invert, backfilling, as-builts, and final cleanup.
- I. Water and Sewer Main Crossings - Sewer Mains: All costs associated with completion of water and sewer main crossings shall be merged with other bid items and will not be considered a separate item for payment. No additional payment will be made for installation of ductile iron pipe associated with water and sewer main crossings.
- J. Concrete Encasement: The concrete encasement shall be measured in linear feet along centerline of the encased pipe, beginning to end of encasement. Payment shall be at the contract unit price shown in the Bid Schedule, which shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for a complete installation.

TP-613 SUBMITTALS:

- A. PVC sewer pipe and fittings
- B. Ductile iron sewer pipe and fittings
- C. Manhole ring and cover
- E. Pre-Cast Manhole including base and pre-cast channels
- F. Sewer main warning tape

- G. Cleanout frame and lid
- H. Exfiltration/Infiltration/Air testing method
- I. Fiberglass gate frame and gates
- J. Detail drawing of size, placement, and spacing of reinforcing bars for manhole bases

TECHNICAL SPECIFICATIONS

SECTION 07 - SEWER SERVICE LINES

TP-700 GENERAL:

The Contractor shall provide all labor, equipment and materials required to install the residence sewer service line indicated on the site layout plans. Installation shall include necessary fittings for connection to the building sewer stubout, tapping of the sewer, installation of the wye or tee as required. All permits, permissions or other authorizations required by the tribal or municipal utility authority for tapping and connection are the responsibility and cost of the Contractor.

TP-701 MATERIALS:

All pipe and fittings required for completion of the sewer service line installation shall meet the requirements of the latest revision of ASTM D 3034. All service line piping shall be 4 inches PVC unless otherwise directed. Pipe shall be nominal size, SDR-35, 0.125-inch minimum wall thickness.

TP-702 INSTALLATION:

Trenching and excavation for sewer service lines shall be in accordance with the provision of Section 01 . The grade from building to sewer main connection shall be uniform and not less than 2 percent. Any changes or deviations in line shall be made with bends not exceeding an angle of 45 degrees.

TP-703 CLEANOUTS:

Sewer service line cleanouts will be installed at the locations indicated in the plans. The cleanout shall be constructed of SDR-35 PVC with a cast iron ferrule equal to a Tyler 2-11. The cleanouts shall be installed with a brass hex socket plug equal to Tyler A Low Square Head cleanout plug. A clear silicon lubricant shall be applied to the cleanout plug to allow for easier removal. The cleanouts shall be constructed in the manner indicated in the detail drawings.

TP-704 SEWER MAIN CONNECTION:

Sewer service line connections to main lines shall be made in accordance with the details as shown on the drawings, or as indicated by the Owner or Owner's Representative. The Contractor shall connect the service line to the main with the appropriate sized sewer saddle or sewer wye as shown on the detail drawings. Sewer wyes shall be PVC. Sewer saddles shall be Romac "CB" or equal. The time and method of connection to existing mains shall be approved by the Owner or Owner's Representative prior to such connection. In no case shall a tapping method be approved that does not provide for a water tight connection to the sewer main.

TP-705 WATER AND SEWER CROSSINGS:

Where water service lines must cross sewer service lines or mains, and the water line is not a minimum of 18 inches above the sewer service line or main, special protection is required. In these instances, the sewer service line or main shall be reconstructed of ductile iron pipe of the same size as the original sewer service line or main for a distance of 10 feet on either side of the water/sewer crossing point. All existing sewer grades shall be maintained.

Water and Sewer Service Line Separation Within 5 ft. of House: This section shall apply to that portion of water and sewer service lines located within five feet of the house. All lines within five feet of the house will be considered as part of the house plumbing. For new construction all service lines shall have 10 foot minimum horizontal separation.

This can best be accomplished by having the water and sewer service lines exit the house 10 feet apart or from different sides. If the 10 foot separation cannot be maintained, and prior written approval is obtained from the Owner or Owner's Representative, and the top of the water service line is at least 12 inches below the bottom of the sewer service line, and the water and sewer service lines are continuous with no joints until the 10 foot separation requirement is met, service lines can be laid closer together than 10 feet.

TP-706 SEPTIC TANK ABANDONMENT

Where shown on the drawings and on the bid schedule, the contractor shall abandon existing septic tank(s) by pumping the tank, knocking a hole in the bottom of the tank to prevent accumulation of water, crush the top of the tank, and backfill with native fill material. The septic tank shall be pumped by a licensed septic tank pumping company and the septage shall be hauled to an approved septage disposal site.

TP-707 AS-BUILT DRAWINGS:

As-built drawings to be furnished by the Contractor for sewer service lines shall include two swing ties from permanent structures or facilities to each of the following:

- A. Tapping point at sewer main
- B. Intersection point with other utilities
- C. Location of cleanouts
- D. Point of connection to the house stubout.

TP-708 MEASUREMENT AND PAYMENT - SEWER SERVICE LINES:

- A. Sewer Service Line: PVC sewer service pipe shall be measured in linear feet along the centerline of the pipe, including fittings. Payment for sewer service lines shall be at the contract unit price shown in the Bid Schedule, this price being full compensation for

furnishing all labor, equipment, materials, and incidentals required for a complete installation; including, excavation, bedding, stabilization material, pipe, sewer saddle or sewer wye, connections to the sewer service cleanout, fittings, trench backfilling, as-builts, and final clean-up.

- B. Cleanouts: Payment for sewer service cleanouts shall be at the contract unit price shown on the Bid Schedule, and shall be full compensation for furnishing all labor, equipment, materials, and incidentals required for complete installation; including, excavation, connections to the tight line and house plumbing, as-builts, and final clean-up.
- C. Septic Tank Abandonment: Payment for abandonment of the existing septic tank shall be at the contract unit price per job and shall be full compensation for pumping and crushing the existing concrete tank, backfilling the void left by the tank, and compacting the soil to prevent subsidence and to bring it to the same level as the surrounding grade. If the existing septic tank is any material other than concrete, the Contractor shall be paid at the contract unit price per job to pump and remove the existing septic tank. The Contractor shall be responsible for all costs associated with the removal, including, but not limited to transporting and disposing of the septic tank at a State licensed and approved waste disposal site located off the particular reservation where the work is to be performed. The payment shall also be full compensation for backfilling the void with soil from the designated area, and compaction to prevent subsidence and to bring it to the same level as the surrounding grade.

TP-709 SUBMITTALS:

All materials listed below will require that a submittal be provided to the Owner for approval prior to the start of any construction requiring those materials.

- A. PVC Sewer Pipe for Service Lines
- B. Service Line Cleanout
- C. Sewer Main Saddle or Wye and Fittings
- D. Permits - if required

TECHNICAL PROVISIONS

SECTION 7 - SEWER SERVICE LINES (MODIFICATION)

MODIFICATION TO TECHNICAL PROVISIONS Technical

Provision Section TP-701 Materials:

Add the following text:

The contractor shall install carsonite marker posts to mark the locations of valves, vaults, bends, tees, crosses, interconnections, cleanouts, water meter boxes, etc. within the right-of-way of roadways. Carsonite markers shall be 72 inches in length, 3.75 inches in width, green in color, model CRM307207 with anchor barb and “Caution Sewer Pipeline” text on the marker as manufactured by Carsonite Composites, or equal and per *Detail Drawing TP-1B Utility Line Marker Detail*. The carsonite markers shall be installed directly over the item that it is marking with an 18 inch bury depth. Carsonite shall be green for sewer system components. Payment for payment will be included in the cost for the system component being marked on this project.